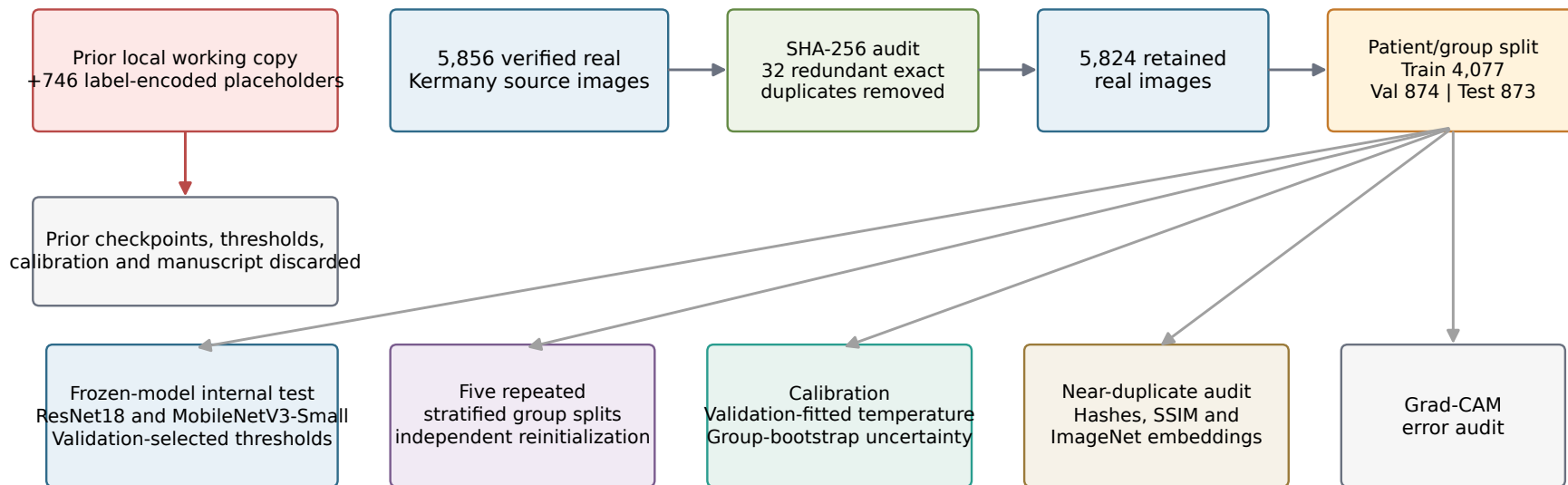
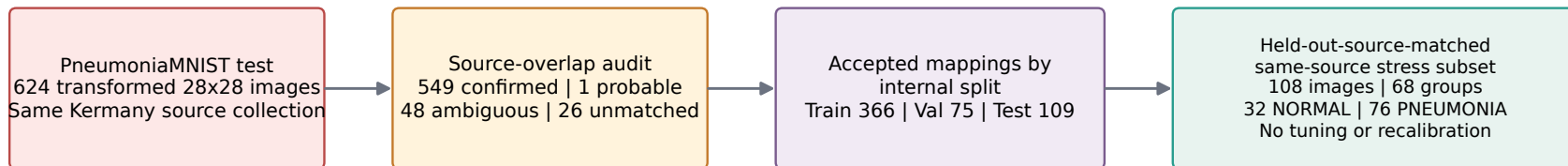


a Data provenance and primary internal evaluation



b Post hoc PneumoniaMNIST provenance and preprocessing-shift analysis



The same-source stress subset evaluates preprocessing and threshold stability, not independent clinical generalization.